

FROM ZERO TO TRADING BOTS

Workshop PINTU x Web3 Dev Bandung

25 Juli 2026 - Parla Cafe, Bandung

Ga perlu jadi programmer. Cukup prompt, download, cron, jalan.

Workshop PINTU x Web3 Dev Bandung

Instructor: Putra

Sabtu, 25 Juli 2026 | 13:00 WIB | Parla Cafe, Bandung

1. Persiapan -- Install Dari Rumah

WiFi venue mungkin lambat. Install ini dari rumah biar ga buang waktu.

1.1 Python 3.10+

Download: <https://www.python.org/downloads/>

Cek setelah install:

```
python --version  
# Harus: Python 3.10.x atau lebih baru
```

PENTING: Centang "Add Python to PATH" pas install!

1.2 curl

Windows 10/11: Sudah include.

Cek:

```
curl --version
```

Mac: Sudah pre-install.

Linux: `sudo apt install curl`

1.3 Code Editor

Pilih salah satu:

VS Code -- <https://code.visualstudio.com/>

Cursor -- <https://cursor.sh/>

Notepad++ -- kalau ringan aja

Step-by-step:

Buka <https://openrouter.ai/keys>

Klik "Sign Up" -- login pake Google / GitHub / email

Setelah login, klik "Create Key"

Nama: `Workshop PINTU` (bebas)

Centang semua permission

Klik "Create" Copy API key yang muncul

Simpan di Notepad -- bakal dipake pas workshop

Kenapa OpenRouter?

Gratis -- model gratis cukup untuk workshop

Ga perlu isi saldo/pulsa

Bisa pake banyak model: DeepSeek, Qwen, Gemini, Llama

Alternatif (kalau gagal daftar OpenRouter):

Gemini API: <https://aistudio.google.com/apikey> -- gratis 60 request/menit

Ollama (local): install + download model gratis, butuh RAM 8GB+

1.5 Install Hermes Agent

```
curl -fsSL https://hermes-agent.nousresearch.com/install.sh | bash
```

Cek:

```
hermes doctor
```

****Catatan Windows:**** Butuh PowerShell sebagai Administrator kalau ada error permission.

1.6 Akun Telegram

Download Telegram di HP: <https://telegram.org/>

Bikin username -- nanti dipake notifikasi bot.

2. Panduan Workshop Lengkap

2.1 Kenapa Bot Sinyal?

Lo ga perlu jadi programmer buat bikin signal bot.

Cukup:

Bisa buka terminal

Bisa copy-paste

Mau nyobain

AI assistant (Hermes Agent) yang nanganin sisanya:

Download template jalanin cron-in sinyal masuk HP

Hasil akhirnya:

Tiap X jam, bot analisa pasar, generate sinyal, kirim ke Telegram.

Lo tinggal baca, eksekusi kalau cocok.

Penting: Ini signal bot, bukan execution bot. Bot kita kasih sinyal beli/jual -- lo sendiri yang eksekusi di exchange. Aman, ga konek API exchange.

2.2 Kenapa Hermes Agent?

Open source -- gratis, ga ada biaya langganan

Cron bawaan -- bot jalan otomatis tanpa server

Multi-provider -- bisa pake Claude, GPT, DeepSeek, bebas

Multi-platform -- Telegram, Discord, file, email

2.3 Install Hermes Agent

Buka terminal (CMD / PowerShell / Terminal Mac/Linux):

```
curl -fsSL https://hermes-agent.nousresearch.com/install.sh | bash
```

Cek:

```
hermes doctor
```

2.4 Setup Provider -- OpenRouter (Gratis)

hermes model

Pilih: OpenRouter
 Paste API key yang udah di-copy tadi
 Pilih model: DeepSeek V3 (gratis, cepet, cukup buat workshop)
 Model gratis lainnya:
Model	Bagus buat
DeepSeek V3	Signal analisa, prompt umum
Qwen 2.5 72B	Coding sederhana
Gemini Flash	Gratis unlimited

2.5 Jalanin Bot Template

Download `trading\_bot\_template.py` dari link workshop.

Jalanin:

```
python trading_bot_template.py
```

Output yang diharapkan:

```

    TRADING SIGNAL -- 25 Jul 13:30 WIB
=====
Price: $4,075.70 (+1.71%)
Direction: LONG | Confidence: 95%
Entry: $4,075.70
SL: $3,962.50 | TP: $4,113.20
R:R 1:1.67

Reasons:
  RSI 66>50 bullish bias
  EMA9>21 bullish
  Price > BB mid
  MACD+ bullish momentum

AI-generated signal. Not financial advice. DYOR.
```

Kalau bot silent (ga ada output)? Itu normal. Bot punya cooldown 4 jam -- ga akan generate sinyal kalau masih ada posisi aktif atau baru aja generate.

Cek history sinyal:

```
cat signals.json
```

Manual `python trading\_bot.py` tiap hari? Capek.

Cron = bot jalan sendiri, lo tinggal terima hasil.

```
hermes cron create 'every 4h' \
--name "Sinyal Auto" \
```

--prompt "Jalankan python trading_bot_template.py 2>&1. Kirim output apa adanya."

Cek cron:

```
hermes cron list
```

2.7 Kirim ke Telegram

```
hermes gateway setup
```

Pilih Telegram

Bikin bot lewat @BotFather di Telegram dapetin token

Start bot di HP

Update cron biar deliver ke Telegram:

```
hermes cron update <job_id> --deliver telegram
```

Tiap 4 jam, sinyal masuk HP. Lo ga perlu buka laptop.

3. Trading Bot Template (Full Code)

File: `trading\_bot\_template.py`

```
#!/usr/bin/env python3
"""
FROM ZERO TO TRADING BOTS -- Workshop PINTU x Web3 Dev Bandung
=====
Bot sederhana: ambil harga hitung indikator generate sinyal.
Ganti `fetch_price()` dengan API real nanti.
"""

import json, math, urllib.request
from datetime import datetime, timezone, timedelta

# Config
COOLDOWN_HRS = 4
MAX_RISK = 100

INDICATORS = {
    "RSI": "Relative Strength Index -- ukur kecepatan perubahan harga",
    "EMA": "Exponential Moving Average -- rata-rata harga terbobot",
    "MACD": "Moving Average Convergence Divergence -- momentum",
    "BB": "Bollinger Bands -- volatility envelope",
    "ATR": "Average True Range -- volatilitas untuk SL/TP",
}

# Price Fetch
def fetch_price():
    """
    TODO: Ganti dengan API real:
    """
```

```

- Kitco: https://www.kitco.com/charts/gold
- Binance: https://api.binance.com/api/v3/ticker/price?symbol=BTCUSDT
- CCXT: library Python untuk 100+ exchange
"""

# Demo data -- diganti peserta nanti
return {"bid": 4075.70, "ask": 4076.10, "change_pct": 1.71, "timestamp": datetime.now().isoformat()}

# Helper
def load_json(path, default):
    try:
        with open(path) as f:
            return json.load(f)
    except (FileNotFoundError, json.JSONDecodeError):
        return default

def save_json(path, data):
    with open(path, "w") as f:
        json.dump(data, f, indent=2)

# Indicator 1: RSI
def calc_rsi(prices, period=14):
    """RSI > 70 = overbought (sinyal SHORT). RSI < 30 = oversold (sinyal LONG)."""
    if len(prices) < period + 1:
        return 50.0
    gains = losses = 0.0
    for i in range(-period, 0):
        d = prices[i] - prices[i - 1]
        if d >= 0:
            gains += d
        else:
            losses -= d
    ag = gains / period
    al = losses / period
    if al == 0:
        return 100.0
    rs = ag / al
    return 100.0 - (100.0 / (1.0 + rs))

# Indicator 2: EMA
def calc_ema(prices, period):
    """EMA9 > EMA21 = bullish (uptrend). EMA9 < EMA21 = bearish (downtrend)."""
    if len(prices) < period:
        return prices[-1] if prices else 0
    k = 2.0 / (period + 1.0)
    ema = sum(prices[-period:]) / period
    for p in prices[-period:]:
        ema = p * k + ema * (1.0 - k)
    return ema

# Indicator 3: MACD
def calc_macd(prices, fast=12, slow=26, signal=9):
    """MACD > 0 = bullish momentum. MACD < 0 = bearish."""
    if len(prices) < slow + signal:

```

```

        return 0.0
    ema_f = calc_ema(prices, fast)
    ema_s = calc_ema(prices, slow)
    return ema_f - ema_s

# Indicator 4: Bollinger Bands
def calc_bollinger(prices, period=20, std_dev=2.0):
    """Upper band = overbought. Lower band = oversold."""
    recent = prices[-period:]
    sma = sum(recent) / period
    var = sum((p - sma) ** 2 for p in recent) / period
    std = math.sqrt(var)
    return sma, sma + std_dev * std, sma - std_dev * std

# Indicator 5: ATR
def calc_atr(history, period=14):
    """ATR besar = market volatil. ATR kecil = market tenang."""
    if len(history) < period:
        return 10.0
    ranges = []
    for i in range(-period, 0):
        h = history[i].get("high", history[i]["bid"])
        l = history[i].get("low", history[i]["bid"])
        ranges.append(h - l)
    return sum(ranges) / period if ranges else 10.0

# Generate Signal
def generate_signal(prices, history, last_signal):
    """
    Combine semua indikator confidence score 0-100%.
    >=60% signal LONG/SHORT.
    <60% skip.
    """

    # Cek cooldown
    if last_signal and last_signal.get("status") == "active":
        return None # masih ada posisi aktif

    if last_signal:
        last_time = datetime.fromisoformat(last_signal["timestamp"])
        elapsed = (datetime.now(timezone.utc) - last_time).total_seconds() / 3600
        if elapsed < COOLDOWN_HRS:
            return None # masih cooldown

    price = prices[-1] if isinstance(prices[-1], (int, float)) else prices[-1].get("bid", prices[-1])
    rsi = calc_rsi(prices)
    ema9 = calc_ema(prices, 9)
    ema21 = calc_ema(prices, 21)
    macd = calc_macd(prices)
    bb_mid, bb_up, bb_lo = calc_bollinger(prices)
    atr = calc_atr(history)

    reasons = []
    conf = 50 # base confidence

```

```

# RSI
if rsi >= 70:
    reasons.append(f"RSI {rsi:.0f}>=70 overbought")
    conf -= 20
elif rsi <= 30:
    reasons.append(f"RSI {rsi:.0f}<=30 oversold")
    conf += 20
elif rsi > 50:
    reasons.append(f"RSI {rsi:.0f}>50 bullish bias")
    conf += 5
else:
    reasons.append(f"RSI {rsi:.0f}<50 bearish bias")
    conf -= 2

# EMA crossover
if ema9 > ema21:
    reasons.append("EMA9>21 bullish")
    conf += 12
else:
    reasons.append("EMA9<21 bearish")
    conf -= 12

# MACD
if macd > 0:
    reasons.append("MACD+ bullish momentum")
    conf += 8
else:
    reasons.append("MACD- bearish momentum")
    conf -= 8

# Bollinger
if price <= bb_lo:
    reasons.append(f"BB lower ${bb_lo:.0f} bounce potential")
    conf += 18
elif price >= bb_up:
    reasons.append(f"BB upper ${bb_up:.0f} rejection potential")
    conf -= 18
elif price > bb_mid:
    reasons.append(f"Price > BB mid ${bb_mid:.0f}")
    conf += 5
else:
    reasons.append(f"Price < BB mid ${bb_mid:.0f}")
    conf -= 2

conf = max(0, min(100, conf))

if conf < 60:
    return None # confidence terlalu rendah

direction = "LONG" if conf > 50 else "SHORT"
sl_dist = atr * 1.5
tp_dist = atr * 2.5

return {
    "timestamp": datetime.now(timezone.utc).isoformat(),
    "direction": direction,

```

```

    "entry": round(price, 2),
    "sl": round(price - sl_dist if direction == "LONG" else price + sl_dist, 2),
    "tp": round(price + tp_dist if direction == "LONG" else price - tp_dist, 2),
    "rr": round(tp_dist / sl_dist, 2),
    "confidence": round(conf),
    "rsi": round(rsi, 1),
    "atr": round(atr, 1),
    "macd": round(macd, 2),
    "bb_mid": round(bb_mid, 2),
    "bb_up": round(bb_up, 2),
    "bb_lo": round(bb_lo, 2),
    "reasons": reasons[:6],
    "status": "active",
    "closed_at": None,
    "exit_price": None,
    "pnl_pct": None,
}

```

Format Output

```

def format_signal(sig, price):
    now = datetime.now(timezone(timedelta(hours=7))).strftime("%d %b %H:%M WIB")
    emoji = "" if sig["direction"] == "LONG" else ""
    lines = [
        f"TRADING SIGNAL -- {now}",
        "=" * 40,
        f"Price: ${price['bid']:.2f} ({price['change_pct']:+.2f}%)",
        f"Direction: {emoji} {sig['direction']} | Confidence: {sig['confidence']}%",
        "",
        f"Entry: ${sig['entry']:.2f}",
        f"SL: ${sig['sl']:.2f} | TP: ${sig['tp']:.2f}",
        f"R:R 1:{sig['rr']}",
        "",
        "Indicators:",
        f"  RSI: {sig['rsi']} | ATR: ${sig['atr']:.1f}",
        f"  BB Mid: ${sig['bb_mid']:.1f} | Upper: ${sig['bb_up']:.1f} | Lower: ${sig['bb_lo']:.1f}",
        f"  MACD: {sig['macd']:+.2f}",
        "",
        "Reasons:",
    ]
    for r in sig["reasons"]:
        lines.append(f"  {r}")
    lines.append("")
    lines.append(" AI-generated signal. Not financial advice. DYOR.")
    return "\n".join(lines)

```

Main Loop

```

def main():
    # 1. Ambil harga
    price = fetch_price()
    if not price:
        print("Gagal ambil harga. Cek koneksi / API.")
        return

    bid = price["bid"]

```

```

# 2. Load history
history = load_json("history.json", [])
candle = {
    "bid": price["bid"],
    "ask": price["ask"],
    "timestamp": datetime.now(timezone.utc).isoformat(),
}
history.append(candle)
if len(history) > MAX_HIST:
    history = history[-MAX_HIST:]
save_json("history.json", history)

prices = [h["bid"] for h in history]

# 3. Load signals
signals = load_json("signals.json", [])

# 4. Cek posisi aktif
active = [s for s in signals if s.get("status") == "active"]
for a in active:
    if a["direction"] == "LONG":
        if bid <= a["sl"]:
            a["status"] = "loss"
            a["closed_at"] = datetime.now(timezone.utc).isoformat()
            a["exit_price"] = bid
            a["pnl_pct"] = round((bid - a["entry"]) / a["entry"] * 100, 2)
            print(f" SL HIT -- Loss ${bid:.2f}")
        elif bid >= a["tp"]:
            a["status"] = "win"
            a["closed_at"] = datetime.now(timezone.utc).isoformat()
            a["exit_price"] = bid
            a["pnl_pct"] = round((bid - a["entry"]) / a["entry"] * 100, 2)
            print(f" TP HIT -- Win ${bid:.2f}")
    else: # SHORT
        if bid >= a["sl"]:
            a["status"] = "loss"
            a["closed_at"] = datetime.now(timezone.utc).isoformat()
            a["exit_price"] = bid
            a["pnl_pct"] = round((a["entry"] - bid) / a["entry"] * 100, 2)
            print(f" SL HIT -- Loss ${bid:.2f}")
        elif bid <= a["tp"]:
            a["status"] = "win"
            a["closed_at"] = datetime.now(timezone.utc).isoformat()
            a["exit_price"] = bid
            a["pnl_pct"] = round((a["entry"] - bid) / a["entry"] * 100, 2)
            print(f" TP HIT -- Win ${bid:.2f}")

save_json("signals.json", signals)

# 5. Generate sinyal baru
last_sig = signals[-1] if signals else None
sig = generate_signal(prices, history, last_sig)

if sig:
    signals.append(sig)
    save_json("signals.json", signals)
    print(format_signal(sig, price))

```

```
else:
    # Bot silent -- normal
    pass

if __name__ == "__main__":
    main()
```

4. Cheatsheet Prompt -- Modifikasi Bot

Template bot udah siap. Tapi kalau mau modifikasi, tanya Hermes Agent pake prompt di bawah. Tinggal copas ganti sesuai kebutuhan. AI akan ubah kodenya.

4.1 Modifikasi Template

Ganti aset (XAU BTC):

Ganti `fetch_price` di `trading_bot_template.py` supaya ambil harga BTC/USD dari CoinGecko API. Pakai url `https://api.coingecko.com/api/v3/simple/price?ids=bitcoin&vs_currencies=usd`. Parse response JSON untuk dapetin harga.

Ubah interval sinyal (cooldown):

Ubah `COOLDOWN_HRS` di `trading_bot_template.py` jadi 2 jam bukan 4 jam.

Tambah indikator baru:

Tambah function `calc_stochastic(prices, period=14)` ke `trading_bot_template.py`. Stochastic oscillator: $%K = (close - lowest\ low) / (highest\ high - lowest\ low) * 100$. Output signal overbought >80 , oversold <20 .

Edit SL/TP rasio:

Ubah perhitungan SL di `trading_bot_template.py` jadi 2x ATR, TP jadi 3x ATR.

Tambah moving average cross sebagai filter:

Tambah filter di `generate_signal()`: pastikan $EMA9 > EMA21$ baru kasih sinyal LONG, kebalikannya buat SHORT.

Ganti bahasa output ke Indonesia:

Ubah `format_signal()` di `trading_bot_template.py` -- ganti semua tulisan ke Bahasa Indonesia. Contoh: 'Direction' 'Arah', 'Confidence' 'Keyakinan'.

Setup cron (bot jalan otomatis):

Bantu saya setup cron job Hermes Agent buat jalanin python `trading_bot_template.py` tiap 6 jam. Kirim output lengkap sinyal ke saya.

Setup Telegram:

Bantu saya setup Hermes Gateway ke Telegram. Saya punya Bot Token dari @BotFather. Langkah-langkahnya?

Kalau ada error, tinggal paste error ke Hermes:

Error ini: [paste error]. Tolong fix.

Error umum:

| Error | Solusi |

|-----|-----|

| ModuleNotFoundError: No module named 'requests' | `pip install requests` |

| SyntaxError | Ada typo di line X |

| Bot jalan tapi ga output | Cooldown masih active / silent mode |

Gunakan ini kapan aja:

"Jelaskan RSI dalam trading dengan analogi sederhana"

"Apa beda EMA sama SMA? Kapan pakai masing-masing?"

"Kenapa SL/TP harus dihitung pakai ATR?"

"Apa itu Bollinger Bands squeeze?"

"Cara backtest sinyal trading tanpa modal?"

"Apa indikator paling penting buat gold trading?"

"Kenapa WR 60% lebih profit daripada WR 90%?"

5. Alur Cepat -- Ringkasan

Setup Provider (Step Detail)

OpenRouter -- Gratis:

Buka <https://openrouter.ai/keys>

Login (Google/GitHub/email)

Klik "Create Key" copy API key

`hermes model` pilih OpenRouter paste key pilih DeepSeek V3

Alternatif:

Gemini API <https://aistudio.google.com/apikey> (60 req/menit gratis)

Ollama local `ollama pull llama3.2` (butuh RAM 8GB+)

Quick Start (6 Langkah)

Install Hermes Agent:

```
curl -fsSL https://hermes-agent.nousresearch.com/install.sh | bash
```

Setup provider:

```
hermes model
```

OpenRouter paste API key DeepSeek V3

Download template:

Dapatkan `trading\_bot\_template.py` dari link workshop

Jalanin bot:

```
python trading_bot_template.py
```

Cron otomatis:

```
hermes cron create 'every 6h' --name "Sinyal Saya" --prompt "Jalankan python trading_bot_template.py 2>&1"
```

Telegram (opsional):

```
hermes_gateway_setup
```

Selesai! Bot sinyal lo udah jalan otomatis.

6. Tips & Best Practices

Mulai 1 aset -- XAU atau BTC, jangan semua
 SL wajib -- jangan entry tanpa SL
 Backtest -- pantau seminggu sebelum real trade
 Cross-check -- AI bisa salah, cek chart manual
 Konsisten -- 3 sinyal bagus > 20 sinyal random
 Jangan langsung trade real money
 Jangan over-trading (cron tiap jam = spam)
 Jangan percaya 100% ke AI -- lo tetap bosnya

7. Resources

Resource	Link
Hermes Agent	https://hermes-agent.nousresearch.com
OpenRouter	https://openrouter.ai/keys
Python Download	https://www.python.org/downloads/
VS Code	https://code.visualstudio.com/
Telegram Bot	https://t.me/botfather
CoinGecko API (data harga gratis)	https://www.coingecko.com/en/api

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